



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

PRE - SCORE

29-10-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	D	D	D	D	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	A,C	A,B,D	D	B,C	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.25 to 1.30	8.45 to 8.50	1.44	32.66 to 32.67	0.56 to 0.58	1256.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	D	C	D	D	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	A,B,D	A,C,D	A,B,C	C,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	8.00	1528.00	17.80	0.30	48.30	2.50

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	B	C	B	A	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C	A,B,C,D	B,C,D	A,B,C,D	A,B	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.00	2.00	1.00	4.00	9.00	3.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

5. Ans (D)

$$r_{\text{rms}} = \sqrt{\frac{\int i^2 dt}{\int dt}}$$

$$i = \frac{2i_0}{T}t$$

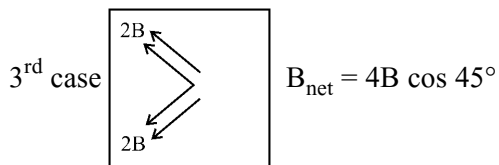
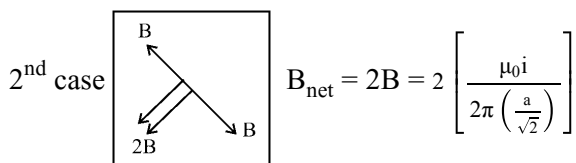
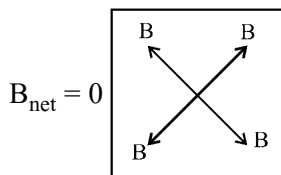
$$= \left[\frac{\frac{4i_0^2}{T^2} \left[\frac{t^3}{3} \right]_0^{T/2}}{\frac{T}{2}} \right]^{1/2} = \frac{i_0}{\sqrt{3}} \text{ (D)}$$

PART-1 : PHYSICS

SECTION-I (ii)

12. Ans (A,B,C)

1st and 4th case



$$= 2\sqrt{2} \left[\frac{\mu_0 i}{2\pi \left(\frac{a}{\sqrt{2}} \right)} \right]$$

PART-1 : PHYSICS

SECTION-II

1. Ans (1.25 to 1.30)

For a ballistic galvanometer, charge $\propto$ deflection

$$\therefore q = VC = K\theta$$

where K is the constant of proportionality

$$\text{or } 2 \times (3 \times 10^{-6}) = K \times 24 \quad \text{or } K = 25 \times 10^{-8}$$

Thus charge-deflection relation for the given

ballistic galvanometer is $q = 25 \times 10^{-8} \theta$

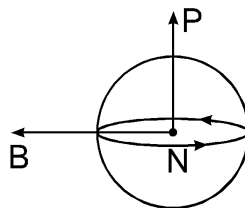
In electromagnetic induction, charge produced

$$= \frac{\text{change of flux}}{\text{resistance}}$$

$$\therefore q = \frac{nSB}{R} = \frac{80 \times (\pi \times 0.5^2 \times 10^{-4}) \times B}{2000} = 25 \times 10^{-8} \times 16$$

$$\Rightarrow B = 1.27 \text{ T}$$

2. Ans (8.45 to 8.50)



Torque on the (coil + sphere) due to flow of charge through coil is

$$= |\vec{p} \times \vec{B}| \quad (\text{where } \vec{p} \text{ is the dipole moment of the coil and } \vec{B} \text{ is the geomagnetic field})$$

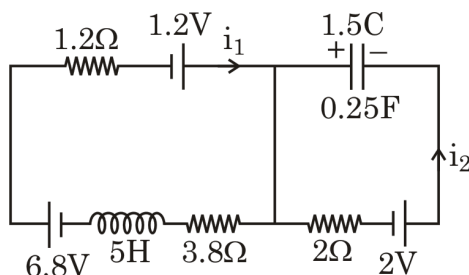
$$= i N \pi r^2 B = I \frac{d\omega}{dt}$$

$$\therefore d\omega = \frac{N \pi r^2 B}{I} i dt$$

$$\text{or } \omega = \frac{N \pi r^2 B}{\frac{2}{3} m r^2} \int_0^{\Delta t} i dt = \frac{3 N \pi B Q}{2} \text{ Ans.}$$

$$\text{Ans: } \omega = \frac{3}{2} \frac{BN\pi Q}{M} = 2.7 \pi \times 10^{-2} \text{ rad/s.}$$

3. Ans (1.44)



Current through key

$$i = i_1 + i_2$$

$$i = \frac{8}{5} (1 - e^{-t}) + \frac{2}{2} (e^{-2t}) + \frac{6}{2} e^{-2t}$$

for minimum current

$$\frac{di}{dt} = 0 \Rightarrow \frac{8}{5} e^{-t} - 8e^{-2t} = 0 \Rightarrow e^{-t} = \frac{1}{5}$$

$$i_{\min} = \frac{32}{25} + \frac{4}{25} = \frac{36}{25} \text{ amp.}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (D)

$$10^{-2} = \frac{10^{-3}}{[\text{OH}^-]^2} \Rightarrow [\text{OH}^-] = 0.316 \text{ M.}$$

2. **Ans (D)**

$$[Aq^+] = \frac{10^{-10}}{0.06} = \frac{10^{-8}}{6}$$

$$[I^-] = \frac{4 \times 10^{-16}}{10^{-8}} \times 6 = 2.4 \times 10^{-7}$$

3. **Ans (C)**

$$\alpha \text{ at } 25^\circ\text{C} = 1.8 \times 10^{-13} \Rightarrow t > 25^\circ\text{C}$$

4. **Ans (D)**

$\Delta S = 0$ for reversibly adiabatic.

5. **Ans (D)**

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} = \Delta H = n C_{Pm} \Delta T$$

6. **Ans (C)**

$$\text{Use } \Delta G = nRT \ln \frac{P_2}{P_1}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (A,B,C)**

(A) For both process, $q = W = \Delta T = \Delta U = \Delta H = 0$

(B) Use $C_m = C_{Vm} + \frac{R}{(1-x)}$

(C) For any process, $\Delta S \geq \int \frac{dq}{T}$

for cyclic $\Delta S = 0$

(D) Reversible adiabatic is isentropic

8. **Ans (A,B,D)**

(A) pH is approximately 3.

(B) $[H_3O^+]_{H_2O} \approx 10^{-11} \Rightarrow \alpha_{H_2O} = 1.8 \times 10^{-13}$

9. **Ans (A,C,D)**

$$4 = pK_b + \log \frac{0.5}{(40M - 0.5)} \quad \dots\dots (1)$$

$$5 = pK_b + \log \frac{2}{(40M - 2)} \quad \dots\dots (2)$$

Solving (1) & (2) we get $pK_b = 4.7$ and

$$M = \frac{3}{40} \text{ molar}$$

10. **Ans (A,B,C)**

(A) $\Delta G = 0$, $q_{rev} \neq q_{irr}$

(B) $\ln K = \frac{-\Delta H^\circ}{R} \times \frac{1}{T} + \frac{\Delta S^\circ}{R}$

(C) At constant P and T, $\Delta G = W_{max}$

11. **Ans (C,D)**

Initially piston is at eq,

$$10^6 = \frac{(2 \times 10^4 + 3 \times 10^4 + m_3) \times 10}{1} + 10^5$$

$$m_3 = \left(\frac{10^6 - 10^5}{10} \right) - 5 \times 10^4 = 10^4 (10 - 1)$$

$$- 5 \times 10^4 = 10^4 (9 - 5) = 4 \times 10^4 \text{ kg}$$

since T = constant

$$P_i V_i = P_B V_B$$

$$10^6 \times 4 = (10^5 + 4 \times 10^5) \times V_B$$

$$V_B = \frac{4 \times 10^6}{10^5 \times 5} = \frac{40}{5} = 8 \text{ litre}$$

Also,

$$P_B V_B = P_C V_C$$

$$10^5 \times 5 \times 8 = 10^5 \times V_C$$

$$V_C = 40 \text{ L}$$

$$W_{BC} = -P_{ext} (V_C - V_B)$$

$$= -10^5 (40 - 8) \times 10^{-3}$$

$$= -3200 \text{ J}$$

12. **Ans (A,B,D)**

S - I $\rightarrow$ S - II isothermal exp
(reversible)

T = constant

V = increases

P = decreases

H = constant (isenthalpic)

S = increases

S - II $\rightarrow$ S - III adiabatic expansion
(reversible)

T = decreases

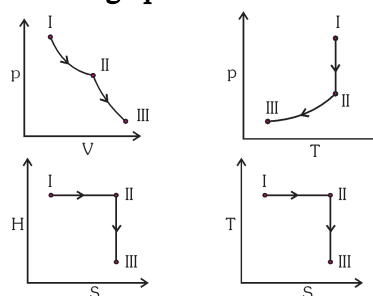
V = increases

P = decreases

H = decreases

S = constant (isentropic)

Correct graphs.



PART-2 : CHEMISTRY

SECTION-II

1. **Ans (8.00)**

$$3\text{I}_2(\text{s}) + 6\text{OH}^- \rightarrow 5\text{I}^-(\text{aq}) + \text{IO}_3^- + 3\text{H}_2\text{O}$$

$$\Delta G^\circ = -250 - 123.5 - 699 + 900 = -172.5$$

$$+ 2.3 \times \frac{25}{3} \times 300 \log \frac{(0.1)^6}{(\text{OH}^-)^6} = 1725$$

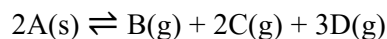
$$\log \frac{0.1}{[\text{OH}^-]} = 5 \Rightarrow \text{pH} = 8$$

2. **Ans (1528.00)**

$$\Delta G = nRT \ln \frac{P_2}{P_1} = 1 \times 8.3 \times 263 \ln \frac{0.16}{0.32}$$

$$= 1528.03 \text{ J.}$$

3. **Ans (17.80)**



at equilibrium $\quad \quad \quad \text{P} \quad \quad 2\text{P} \quad \quad 3\text{P}$
as initially there was only A(s) so pressure will be in ratio of moles.

$$\text{Given } P + 2P + 3P = 6P = 12 \quad \boxed{P = 2 \text{ bar}}$$

$$K_p = P \times (2P)^2 \times (3P)^3$$

$$= 2 \times (4)^2 \times (6)^3$$

$$= 32 \times 8 \times 27$$

$$K_p = 2^8 \times 3^3$$

$$\ln K_p = 8 \ln 2 + 3 \ln 3$$

$$= 8 \times 0.7 + 3 \times 1.1$$

$$= 5.6 + 3.3 = 8.9$$

$$\Delta G^\circ = -RT \ln K_p$$

$$|\Delta G^\circ| = 2 \times 1000 \times 8.9 = 17.8 \text{ Kcal.}$$

4. **Ans (0.30)**

$$a_{\text{FCC}} = 2\sqrt{2}R = 2\sqrt{2} \times 100\sqrt{3} \times 10^{-8} = 2\sqrt{6} \times 10^{-6} \text{ cm.}$$

$$a_{\text{FCC}}^3 = (2\sqrt{6} \times 10^{-6})^3 = (5 \times 10^{-6})^3 = 125 \times 10^{-18} \text{ ml.}$$

$$\therefore V_1 = 125 \times 10^{-18} \times 10^{15} = 125 \times 10^{-3} \text{ ml.}$$

$$a_{\text{bcc}} = \frac{4R}{\sqrt{3}} = \frac{4 \times 100\sqrt{3} \times 10^{-8}}{\sqrt{3}} = 4 \times 10^{-6} \text{ cm.}$$

$$a_{\text{bcc}}^3 = (4 \times 10^{-6})^3 = 64 \times 10^{-18} \text{ ml.}$$

$$\therefore V_2 = 64 \times 10^{-18} \times 10^{15} \times 2 = 128 \times 10^{-3} \text{ ml.}$$

$$\Delta V = V_2 - V_1 = (128 - 125) \times 10^{-3} = 3 \times 10^{-3}$$

$$W = -P_{\text{ext}} \Delta V = -1000 (3 \times 10^{-3}) = -3 \text{ bar ml} = -0.3 \text{ J}$$

Ans. = 0.3

5. **Ans (48.30)**

$$\Delta G^\circ = -2.3 RT \log \frac{P_{\text{O}_3}^2}{P_{\text{O}_2}^3}$$

6. **Ans (2.50)**

$$[\text{Br}^-] = [\text{Cl}^-] = \frac{0.02 \times 250}{1000} = 5 \times 10^{-3}.$$

(Initially)

$$[\text{Ag}^+] = 0.005 = 5 \times 10^{-3}. \text{ (Initially)}$$

AgBr will be deposited almost completely.

$$[\text{Ag}^+] = \frac{K_{\text{SP AgCl}}}{[\text{Cl}^-]} = \frac{10^{-10}}{5 \times 10^{-3}} = \frac{10^{-7}}{5}$$

$$[\text{Br}^-] = \frac{K_{\text{SP AgBr}}}{[\text{Ag}^+]} = \frac{5 \times 10^{-13}}{10^{-7}} \times 5$$

$$= 25 \times 10^{-6} = 2.5 \times 10^{-5}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**

Let ΔABC have vertices

$$A(a_1t_2, a(t_1 + t_2))$$

$$B(a_2t_3, a(t_2 + t_3))$$

$$C(a_3t_1, a(t_3 + t_1))$$

$$m_{BC} = \frac{1}{t_3} \Rightarrow m_{AD} = -t_3$$

$\Rightarrow$ Equation of AD is

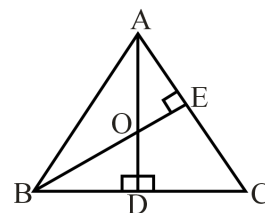
$$t_3x + y = a_1t_2t_3 + a(t_1 + t_2) \quad \dots(1)$$

Similarly Equation of BE is

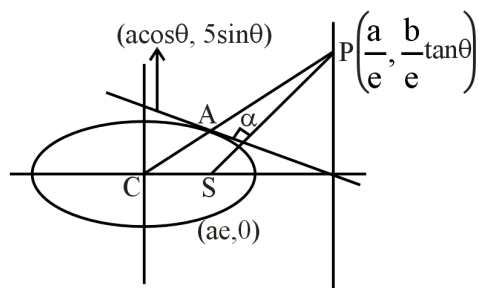
$$t_1x + y = a_1t_2t_3 + a(t_2 + t_3) \quad \dots(2)$$

by solving (1) & (2) we get

$$0, (-a, a(t_1t_2t_3 + t_1 + t_2 + t_3))$$



2. Ans (B)



Let the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$A = (a \cos \theta, b \sin \theta)$$

Equation of AC will be $y = \frac{b}{a} \tan \theta x$

Solving with $x = \frac{a}{e}$, we get $P\left(\frac{a}{e}, \frac{b}{e} \tan \theta\right)$

Slope of tangents at A is $-\frac{b}{a \tan \theta}$

$$\text{Slope of PS} = \frac{\frac{b}{e} \tan \theta}{\frac{a}{e} - ae} = \frac{b \tan \theta}{a(1 - e^2)} = \frac{a}{b} \tan \theta$$

$$\Rightarrow \alpha = \frac{\pi}{2}$$

3. **Ans (C)**

$$T \equiv \frac{x \cos \theta}{3} + \frac{y \sin \theta}{2} = 1$$

$$x = 0, y = 2\operatorname{cosec}\theta$$

chord A'P :

$$y = \frac{2 \sin \theta}{3(\cos \theta + 1)}(x + 3)$$

$$x = 0, OM = \frac{2 \sin \theta}{(\cos \theta + 1)}$$

$$\text{Now, } OQ^2 - MQ^2 = OQ^2 - (OQ - OM)^2$$

$$= 2(OM)(OQ) - OM^2$$

$$= OM[2OQ - OM]$$

$$= \frac{2 \sin \theta}{(\cos \theta + 1)} \left[\frac{4}{\sin \theta} - \frac{2 \sin \theta}{1 + \cos \theta} \right]$$

$$= 2 \sin \theta [4 + 4 \cos \theta - 2 \sin^2 \theta]$$

$$= \frac{(1 + \cos \theta)^2 \sin \theta}{4 \sin \theta [2 + 2 \cos \theta - 1 + \cos^2 \theta]}$$

$$= 4$$

4. Ans (B)

Let equation of hyperbola be $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$;

$$b^2 = a^2(e^2 - 1)$$

Let P be $(a \sec \theta, b \tan \theta)$

Equation of normal $a \cos \theta + b \cot \theta = a^2 + b^2$

$$L \equiv \left(\frac{a^2 + b^2}{a} \sec \theta, 0 \right)$$

$$\mathbf{M} \equiv \left(0, \frac{a^2 + b^2}{a} \tan \theta \right)$$

Mid point of LM $\equiv (h, k)$

$$\equiv \left(\frac{a^2 + b^2}{2a} \sec \theta, \frac{a^2 + b^2}{2b} \tan \theta \right)$$

$$\frac{2ah}{a^2 + b^2} = \sec \theta \quad \& \quad \tan \theta = \frac{2bk}{a^2 + b^2}$$

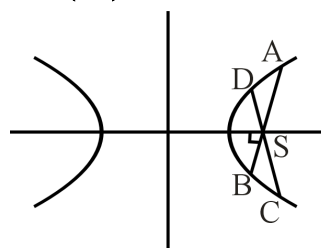
$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\frac{4a^2h^2}{(a^2 + b^2)^2} - \frac{4b^2k^2}{a^2 + b^2} = 1$$

$$e' = \sqrt{1 + \frac{4a^2}{4b^2}} = \sqrt{1 + \frac{1}{e^2 - 1}}$$

$$e' = \sqrt{\frac{e^2}{e^2 - 1}}$$

5. **Ans (A)**



$$\frac{x-1}{\cos \theta} = \frac{y-0}{\sin \theta} = r$$

$$9(r\cos\theta + 1)^2 - r^2\sin^2\theta = 1$$

$$r^2(9\cos^2\theta - \sin^2\theta) + 18r\cos\theta + 8 = 0$$

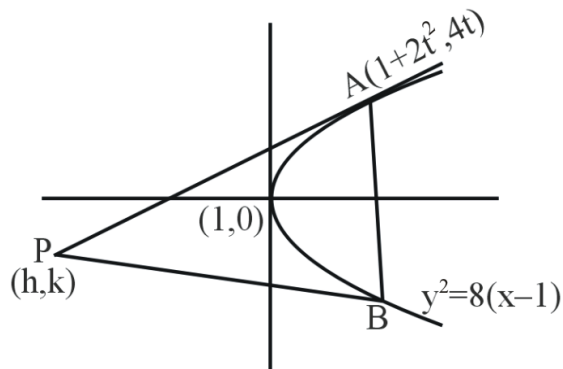
$$\frac{1}{r_1 r_2} = \frac{9 \cos^2 \theta - \sin^2 \theta}{8}$$

$$\frac{1}{r_3 r_4} = \frac{9 \sin^2 \theta - \cos^2 \theta}{8}$$

$$\frac{1}{r_1 r_2} + \frac{1}{r_3 r_4} = 1$$

6. Ans (B)

$$B \left(1 + 2 \left(-t - \frac{2}{t} \right)^2, 4 \left(-t - \frac{2}{t} \right) \right)$$



$$h = 1 + 2t \left(-t - \frac{2}{t} \right)^2 \quad k = 2 \left(t - t - \frac{2}{t} \right)$$

$$\Rightarrow t = -\frac{4}{k}$$

$$h = 1 - \frac{8}{k} \left(\frac{4}{k} + \frac{k}{2} \right) \Rightarrow y^2(x+3) = -32$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A,C)

Equation of chord of contact

$$\alpha x + \beta y = 4$$

...(1)

Also, Equation of tangent at P(t)

$$\text{is } ty = x + 2t^2$$

$$\text{Also } t\beta = \alpha + 2t^2$$

from (1) & (2)

$$(t\beta - 2t^2)x + \beta y = 4$$

$$\beta(tx + y) - (2t^2x + 4) = 0$$

$$\therefore tx + y = 0 \text{ \& } 2t^2x + 4 = 0$$

$$t = -\frac{y}{x}$$

$$\therefore 2 \frac{y^2}{x^2} \times x + 4 = 0$$

$$y^2 + 2x = 0$$

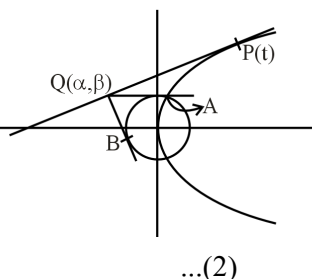
Area bounded by S = 0 &

the line $2x + 1 = 0$

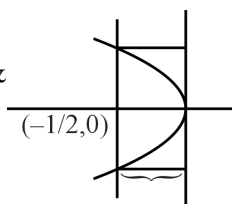
$$\frac{2}{3} \times \frac{1}{2} \times 2 = \frac{2}{3}$$

$2x - y + 1 = 0$ is a focal chord.

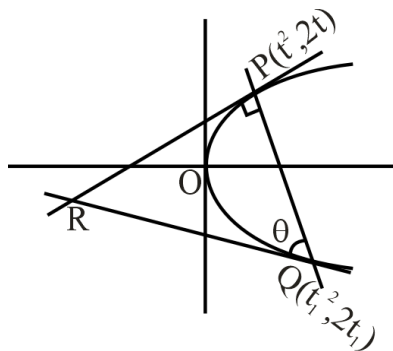
Hence angle is 90°



...(2)



8. Ans (A,B,C,D)



Let $P(x_0, y_0)$ be $(t^2, 2t)$

$$\Rightarrow x = t^2, y_0 = 2t$$

$$t_1 = -t - \frac{2}{t}$$

$$\Rightarrow tt_1 = -t^2 - 2$$

$$\tan \theta = \left| \frac{-t - \frac{1}{t_1}}{1 + (-t) \cdot \frac{1}{t_1}} \right| = \left| \frac{tt_1 + 1}{t_1 - t} \right| = \frac{|t|}{2} = \frac{|y|}{4}$$

Length of normal chord

$$PQ = \sqrt{(t^2 - t_1^2)^2 + 4(t - t_1)^2}$$

$$= |t - t_1| \sqrt{(t + t_1)^2 + 4}$$

$$= \left| 2t + \frac{2}{t} \right| \sqrt{\frac{4}{t^2} + 4}$$

$$= \frac{4(t^2 + 1)^{3/2}}{t^2} = \frac{4(x_0 + 1)^{3/2}}{x_0}$$

Coordinates of R are $(tt_1, t + t_1)$

$$\Rightarrow \text{ordinate of R is } t + t_1 = -\frac{2}{t} = -\frac{4}{y_0}$$

$\therefore$ Triangle PQR is right angled.

$\therefore$ its circumcentre is mid point of QR

$$\Rightarrow \text{its ordinate is } \frac{t + t_1 + 2t_1}{2} = \frac{-2(x_0 + 3)}{y_0}$$

9. Ans (B,C,D)

$P(4, 3) \quad 4x - 3y = 7$

$$\frac{x-4}{\frac{3}{5}} = \frac{y-3}{\frac{4}{5}} = -\frac{5}{2} \Rightarrow Q\left(\frac{5}{2}, 1\right)$$

$$\frac{x-4}{\frac{3}{5}} = \frac{y-3}{\frac{4}{5}} = -5 \Rightarrow R(1, -1)$$

Let $S_1 : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

& $S_2 : \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1, A < B$

for $S_1 : \frac{5x^2}{2a^2} - \frac{y^2}{b^2} = 1$

$$\Rightarrow \frac{5}{2a^2} = \frac{4}{7}; \frac{1}{b^2} = \frac{3}{7}$$

$$S_1 : \frac{x^2}{35/8} - \frac{y^2}{7/3} = 1$$

$$\Rightarrow e_1^2 : \frac{7/3}{35/8} + 1 \Rightarrow e_1^2 = \frac{23}{15}$$

for $S_2 : \frac{x}{A^2} - \frac{y}{B^2} = 1 \Rightarrow A^2 = \frac{7}{4}, B^2 = \frac{7}{3}$

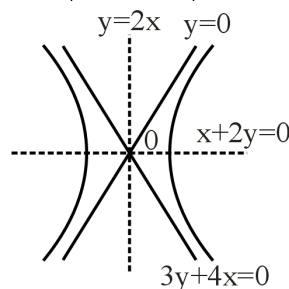
$$\frac{x^2}{7/4} + \frac{y^2}{7/3} = 1 \Rightarrow e_2^2 = 1 - \frac{7/4}{7/3}$$

$$\Rightarrow e_2^2 = \frac{1}{4} \Rightarrow e_2 = \frac{1}{2}$$

$$S_1 \& S_2 \Rightarrow x^2 \left(\frac{4}{7} + \frac{8}{35} \right) = 2 \Rightarrow y \notin \mathbb{R}$$

$\Rightarrow$ 4 common tangents

10. Ans (A,B,C,D)



$y = 0, 3y + 4x = 0$ are asymptotes whose angle bisector are

$$y = 2x, x + 2y = 0$$

solving H with $x + 2y = 0$

$$y^2 = \frac{4}{5} \Rightarrow y = \pm \frac{2}{\sqrt{5}}$$

so vertex $\left(-\frac{4}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right) \left(\frac{4}{\sqrt{5}}, -\frac{2}{\sqrt{5}}\right)$

C.A : $y = 2x, a = 2, b = 1$

$$\text{OR H : } \frac{\left(\frac{2x-y}{\sqrt{5}}\right)^2}{4} - \frac{\left(\frac{x+2y}{\sqrt{5}}\right)^2}{1} = 1$$

$$L = 1, e^2 = \frac{5}{4}$$

11. Ans (A,B)

$$P_1 : y^2 = \ell_1 x$$

$$P_2 : y^2 = -\ell_2(x - b)$$

point on $P_1 \left(\frac{y^2}{\ell_1}, y\right)$ and $P_2 \left(b - \frac{y^2}{\ell_2}, y\right)$

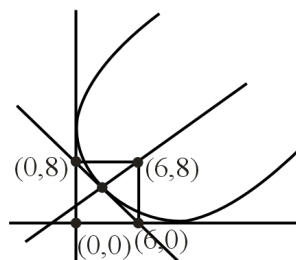
find locus of midpoint of above points

12. Ans (A,B,C)

A perpendicular dropped from focus to any tangent

meet the tangent at the vertex circle is $(x - 6) x +$

$$y(y - 8) = 0 \Rightarrow x^2 - 6x + y^2 - 8y = 0$$



$$\text{Equation of directrix is } y = \frac{-4}{3}x$$

$$\therefore \text{distance of } (6, 8) \text{ from the directrix is } 2a = \frac{48}{5}$$

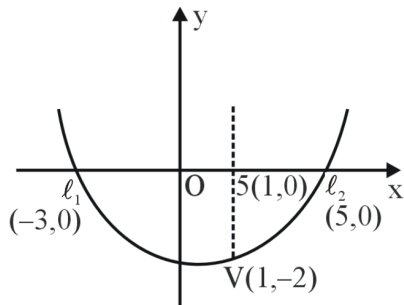
$$\Rightarrow a = \frac{24}{5}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (2.00)

Let $y = mx + c$ chord



$$\therefore 3x^2 - y^2 - (2x - 4y) \left(\frac{y - mx}{c} \right) = 0$$

coeff of x^2 + coeff of $y^2 = 0$

$$(2x + y) + c(x - 1) = 0.$$

$\Rightarrow$ fixed point vertex is $(1, -2)$

$$\text{and } 4a = 8 = L_1 L_2$$

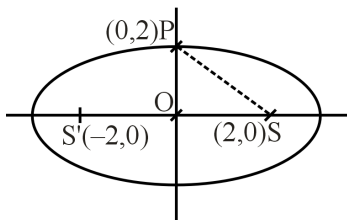
$$\Rightarrow a = 2$$

$$\therefore L_1 s = L_2 S = 4$$

$$L_1(-3, 0) \text{ \& } L_2(5, 0)$$

$$\therefore (\alpha + \beta + \gamma + \delta) = 2$$

2. Ans (2.00)



$$e = \sqrt{1 - \frac{4}{8}} = \frac{1}{\sqrt{2}}$$

If P, S are ends of LR : New focus $(1, 1)$

Directrix will be parallel to PS and pass through $(-1, -1)$.

$$\therefore \text{Let directrix : } x + y = \lambda$$

$$\Rightarrow -1 - 1 = \lambda \Rightarrow \lambda = -2$$

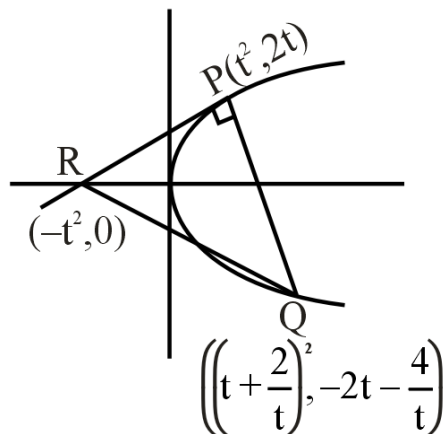
$$\therefore \text{Directrix} \equiv x + y + 2 = 0$$

$\therefore$ Required equation :

$$\sqrt{(x-1)^2 + (y-1)^2} = \left| \frac{x+y+2}{\sqrt{2}} \right|$$

$$\Rightarrow 2(x-1)^2 + 2(y-1)^2 = (x+y+2)^2$$

3. Ans (1.00)



$$3h = \left(t + \frac{2}{t} \right)^2; 3k = 2t - 2t - \frac{4}{t}$$

$$\Rightarrow t = -\frac{4}{3k}$$

$$\therefore 3h = \left(-\frac{4}{3k} + \frac{2.3k}{-4} \right)^2$$

$$= 3h = \frac{(8 + 9k^2)^2}{36k^2}$$

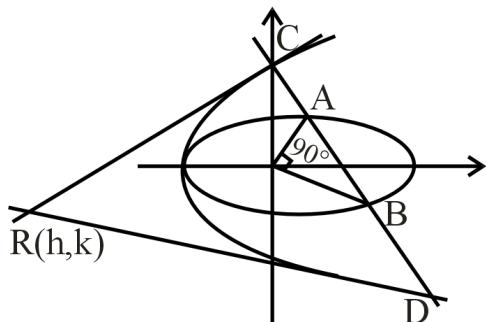
$$\Rightarrow 3 \times 36hk^2 = (8 + 9k^2)^2$$

$$\Rightarrow 3 \times 36xy^2 = (8 + 9y^2)^2$$

$$\Rightarrow 108xy^2 = (8 + 9y^3)^2$$

$$b - a = 1$$

4. Ans (4.00)



$$y = px + q \text{ (AB)} \quad \dots(\text{i})$$

Homogenising 'E'

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \left(\frac{y - px}{q} \right)^2$$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} = \frac{1+p^2}{q^2}$$

$$\Rightarrow q^2(a^2 + b^2) = a^2b^2(1 + p^2) \quad \dots(ii)$$

Chord of contact of parabola

$$y_k = 4c \left(\frac{x+h}{2} + a \right)$$

$$\Rightarrow y_k = 2cx + 2ch + 4ca \quad \dots(\text{iii})$$

comparing (i) and (iii)

$$\frac{1}{k} = \frac{p}{2c} = \frac{q}{2ch + 4ca}$$

put in (ii)

$$\therefore a^2b^2(y^2 + 4c^2) = 4c^2(x + 2a)^2(a^2 + b^2)$$

$$\therefore \alpha = 4, \beta = 2$$

5. Ans (9.00)

Let foot of perpendicular from $S(-1, -1)$ on

 $3x - y = 8$ is (α, β)

$$\therefore \frac{\alpha+1}{3} = \frac{\beta+1}{-1} = 1 \quad \therefore (\alpha, \beta) = (-2, -2)$$

$\Rightarrow$ point of intersection of axis and tangent is

$(-3, -17)$

$\therefore$ slope of axis is $= 8$

Image of S about directrix is S' (5, -3)

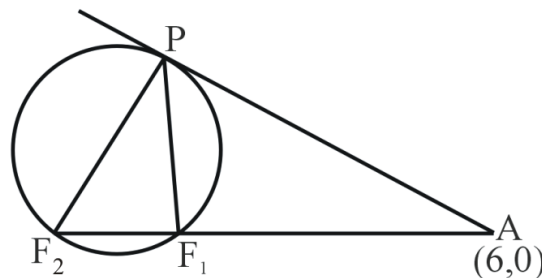
$\therefore$ Equation of directrix is $8y + x + 19 = 0$

$$\Rightarrow a + b = 9$$

6. Ans (3.00)

$\angle F_1PF_2$ is maximum if P is point of tangency of

circle through F_1, F_2 & P F_2 on $y + 2x = 12$



$$\angle APF_1 = \angle AF_2P \Rightarrow \triangle APF_1 \sim \triangle AF_2P$$

$$\Rightarrow \frac{F_1P}{F_2P} = \frac{AP}{AF_2} \text{ \& } AP^2 = AF_1AF_2$$

$$\Rightarrow \left(\frac{PF_2}{PF_1} \right)^2 = \frac{AF_2}{AF_1} = 3$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

PRE - SCORE

29-10-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,C,D	B,D	B,D	A,C,D	B,D	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	4.50	3000.00	1.89 to 1.91	78.12 to 78.13	10.00	10.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	9	9	9	9	2

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,C	C,D	A,B,D	A,C	A,B,C	C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.50	369.00	100.00	16.00	375.00	750.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	1	6	2	2	4

PART-3 : MATHEMATICS

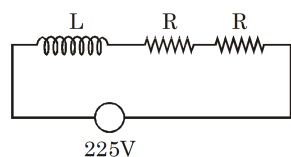
SECTION-I	Q.	1	2	3	4	5	6
	A.	B,D	A,C	B,C	A,B,C	B,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	42.66 or 42.67	72.00	16.00	38.00	32.00	50.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	6	4	6	5	7	2

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

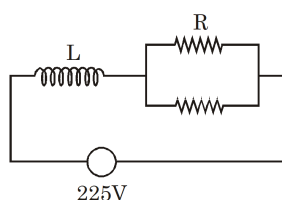
1. Ans (A,B,C,D)



$$P_1 = V_{\text{rms}} i_{\text{rms}} \cos \phi$$

$$P_1 = \frac{V_{\text{rms}}^2}{Z} \cdot \frac{R_{\text{eq}}}{Z} = \frac{(225)^2 \cdot 2R}{X_L^2 + (2R)^2}$$

$$810 = \frac{(225)^2 \cdot 2R}{X_L^2 + 4R^2} \dots (i)$$



$$P_2 = V_{\text{rms}} i_{\text{rms}} \cos \phi$$

$$P_2 = \frac{V_{\text{rms}}^2}{Z} \cdot \frac{R_{\text{eq}}}{Z} = \frac{(225)^2 (R/2)}{X_L^2 + \left(\frac{R}{2}\right)^2}$$

$$506.25 = \frac{(225)^2 R}{2 \left(X_L^2 + \frac{R^2}{4} \right)} \dots (ii)$$

From (i) & (ii), we get

$$R = 20 \, \Omega, X_L = 30 \, \Omega$$

$$\tan \phi_1 = \frac{X_L}{R_{\text{eq}}} = \frac{X_L}{2R} = \frac{30}{40}$$

$$\phi_1 = 37^\circ$$

$$\& \tan \phi_2 = \frac{X_L}{R_{\text{eq}}} = \frac{X_L}{\left(\frac{R}{2}\right)} = \frac{2X_L}{R}$$

$$\tan \phi_2 = \frac{60}{20} = 3$$

$$\phi_2 = \tan^{-1}(3)$$

2. Ans (B,D)

Torque of magnetic field : $\tau = MB_0 = Ia^2B_0$

For rotational equilibrium : $\tau = mgx$

$$\Rightarrow Ia^2B_0 = mgx$$

$$\Rightarrow x = \frac{Ia^2B_0}{mg}$$

For the block not to topple :

$$x < \frac{a}{2} \Rightarrow I = \frac{mg}{2aB_0}$$

3. Ans (B,D)

Induced emf in the loops,

$$\varepsilon = \frac{R^2\theta}{2} \frac{dB}{dt} = \frac{a_0R^2\theta}{2}$$

Induced current in the loop,

$$I = \frac{\varepsilon}{2R(1 + \cos\theta)\lambda} = \frac{a_0R\theta}{4\lambda(1 + \cos\theta)}$$

$$V_O - V_Q = I\lambda 2R \cos\theta = \frac{a_0R^2\theta \cos\theta}{2(1 + \cos\theta)}$$

At $\theta = 45^\circ$,

$$V_O - V_Q = \frac{a_0R^2\pi}{8(\sqrt{2} + 1)}$$

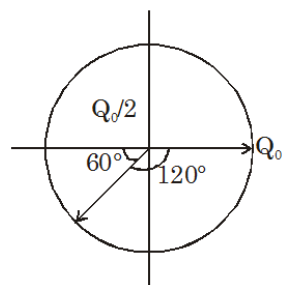
$$V_P - V_Q = e - I\lambda R = \frac{a_0R^2\theta}{2} - \frac{a_0R^2\theta}{4(1 + \cos\theta)}$$

$$= \frac{a_0R^2\theta}{4} \left(\frac{1 + 2\cos\theta}{1 + \cos\theta} \right)$$

For $\theta = 45^\circ$

$$V_P - V_Q = \frac{a_0R^2\pi}{8\sqrt{2}}$$

4. Ans (A,C,D)



$$L_{eq} = \frac{2L \times L}{2L + L} = \frac{2L}{3}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{2LC}{3}}$$

$$t = \frac{2\pi}{3}\sqrt{\frac{2LC}{3}}$$

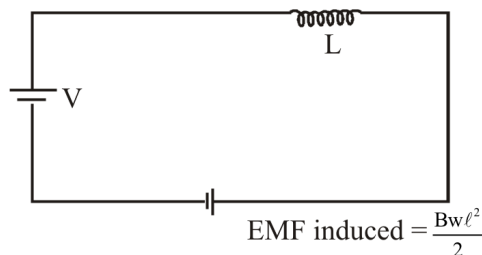
$$i_{max} = Q_0 \times \omega = Q_0\sqrt{\frac{3}{2LC}}$$

$$i_i = \frac{2}{3} = Q_0\sqrt{\frac{3}{2LC}} = Q_0\sqrt{\frac{2}{3LC}}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (4.50)



$$V - L \frac{di}{dt} - \frac{B\omega\ell^2}{2} = 0 \dots (1)$$

$$-L \frac{d^2i}{dt^2} - \frac{B\ell^2}{2} \cdot \frac{3}{2} \frac{Bi}{M} = 0$$

$$\frac{d\omega}{dt} \cdot \frac{M\ell^2}{3} = \frac{B \cdot i\ell^2}{2}$$

$$\Rightarrow \frac{d\omega}{dt} = \frac{3}{2} \frac{Bi}{M}$$

$$\frac{d^2i}{dt^2} = -\frac{3B^2\ell^2}{4ML} i$$

on solving

$$\Rightarrow i = i_0 \sin(\Omega t) \text{ where } \Omega = \sqrt{\frac{3B^2\ell^2}{4ML}}$$

$$\text{now } \frac{d\omega}{dt} = \frac{3}{2} \frac{B}{M} \cdot i_0 \sin(\Omega t)$$

$$\omega = \int_0^t \frac{3}{2} \frac{B}{M} i_0 \sin(\Omega t) dt$$

$$\omega = \int_0^t \frac{3}{2} \frac{B}{M} i_0 \sin(\Omega t) dt = \frac{3}{2} \frac{Bi_0}{M} \frac{1}{\Omega} (-\cos(\Omega t))_0^t$$

$$\omega = \frac{3B}{2M\Omega} i_0 (1 - \cos(\Omega t))$$

But at $t = 0$ $\omega = 0$

by eq. (1)

$$L \frac{di}{dt} = V - \frac{B\omega\ell^2}{2}$$

at $t = 0$

$$L i_0 \cos(\Omega t) \Omega = V - \frac{B\ell^2}{2} (0)$$

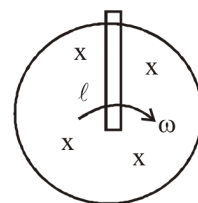
$$\text{or } Li_0\Omega = V$$

$$\Rightarrow i_0 = \frac{V}{L\Omega} = \frac{V}{L} \cdot \sqrt{\frac{4ML}{3B^2\ell^2}} = \frac{2V}{B\ell} \sqrt{\frac{M}{3L}}$$

$$\Rightarrow \omega_{max} = \frac{3B}{2M\Omega} \cdot \frac{V}{L\Omega} = \frac{3BV}{2ML} \cdot \left(\frac{3B^2\ell^2}{4ML} \right) = \frac{9B^3\ell^2V}{8M^2L^2}$$

Put $B = 1T$, $\ell = 1m$, $V = 4 \text{ volt}$

$$M = 1 \text{ kg}, L = 1H \Rightarrow \omega_{max} = \frac{9}{2} = 4.5 \text{ rad/sec.}$$



3. **Ans (1.89 to 1.91)**

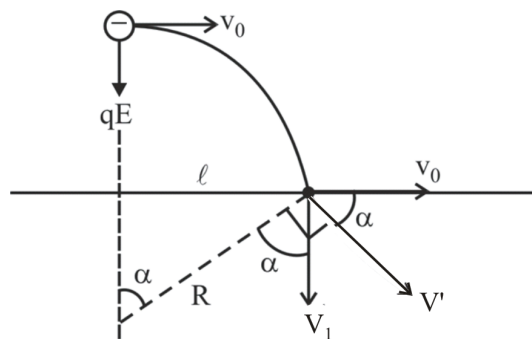
Without loss of generality, let the current around A flow in the counterclockwise direction and let the flux through A be positive. Note that the current will flow in the clockwise direction in C and around B, the area enclosed by the loop is B + C. The flux will be negative here. Therefore, the total flux is then proportional to:

$$\Phi \propto A - B - 2C = 0.81$$

and the magnitude of the induced electromotive force is:

$$\varepsilon = (A - B - 2C) \frac{dB}{dt} = 1.9 \text{ T m}^2 \text{ s}^{-1}$$

6. **Ans (10.00)**



$$\sin \alpha = \frac{v_1}{v_0} = \frac{l}{R}$$

$$\frac{\frac{qE}{m} \cdot t}{v_1} = \frac{v_0 t}{\frac{mv_1}{qB}} = \frac{v_0 t q B}{m}$$

$$E = v_0 B \Rightarrow B = \frac{E}{v_0}$$

PART-1 : PHYSICS

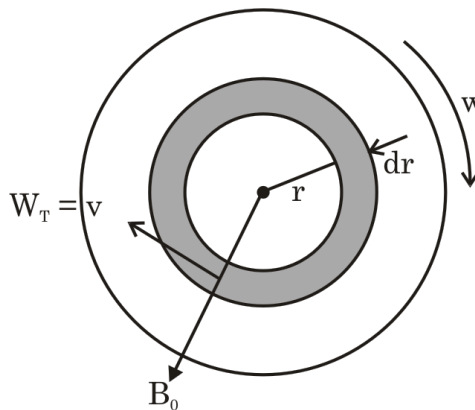
SECTION-III

1. **Ans (3)**

$$\begin{aligned} v_1 &= -L_1 \frac{dI_1}{dt} - M \frac{dI_2}{dt} \\ &= -(0.050)(120) - (0.015)(-200) = -3V \end{aligned}$$

3. **Ans (9)**

The force on any small part of the disc is in the vertically upward direction.



$$dF = \left(\frac{Q}{\pi R^2} 2\pi r dr \right) \omega r B_0$$

$$dF = \frac{2Q\omega B_0}{R^2} r^2 dr$$

$$F = \frac{2}{3} Q\omega B_0 R = Mg$$

$$\omega = \frac{3Mg}{2QB_0 R} = \frac{3 \times 2 \times 10}{2 \times 2 \times 10^{-2} \times 10 \times \frac{1}{6}} = 9 \times 10^2 \text{ rad/s}$$

PART-2 : CHEMISTRY

SECTION-I

1. **Ans (A,B,C)**

Buffer solution will never be formed by mixing the solutions of strong acid and strong base.

2. **Ans (C,D)**

Heat capacity and G are extensive property.

3. **Ans (A,B,D)**

$$\Delta_r H^0 = 2(-110) - (-393) = 173 \text{ kJ/mol}^{-1}$$

$$\Delta_r S^0 = 2(200) - 214 - 6 = 180 \text{ J/mol}^{-1} \text{ K}^{-1}$$

$$\Delta_r G^0 = 173 - \frac{300 \times 180}{1000} = 119 \text{ kJ/mol}$$

$$= -RT \ln K_p \Rightarrow K_p < 1$$

4. **Ans (A,C)**

$$s = [\text{Zn}^{2+}] + [\text{Zn}(\text{OH})_4]^{2-}$$

$$\text{or, } s = \frac{10^{-17}}{[\text{OH}^-]^2} + 0.1 [\text{OH}^-]^2$$

$$\text{For } s \text{ to be minimum, } \frac{ds}{d[\text{OH}^-]} = 0$$

$$\Rightarrow [\text{OH}^-] = 10^{-4} \text{ M.}$$

5. Ans (A,B,C)

At 200 K,

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$

$$\text{or, } 20 = \Delta H^\circ - \frac{200 \times (-20)}{1000}$$

$$\text{or, } \Delta H^\circ = 16 \text{ kJ mol}^{-1}$$

$$\Delta H_{T_2}^\circ = \Delta H_{T_1}^\circ + \Delta C_p (T_2 - T_1) \rightarrow \text{Kirchoff's equation.}$$

$$\text{or, } \Delta H_{400 \text{ K}}^\circ = \Delta H_{200 \text{ K}}^\circ + 20(400 - 200)$$

$$\text{or, } \Delta H_{400 \text{ K}}^\circ = 16 + \frac{20 \times 200}{1000} = 20 \text{ kJ mol}^{-1}.$$

6. Ans (C,D)

$$8.92 = 7 + \frac{1}{2} (\text{pKa} + \log 0.1) \Rightarrow \text{pKa} = 4.84$$

$$8.92 = 9.52 + \log \frac{[\text{In}]}{[\text{HIn}]} \Rightarrow \frac{[\text{In}]}{[\text{HIn}]} = \frac{1}{4}$$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (1.50)

$$[\text{HC}_4\text{H}_4\text{O}_6^-] = \sqrt{K_a \cdot C} = \sqrt{10^{-3} \times 1.0} = 0.0316 \text{ M}$$

$$\therefore [\text{K}^+] = \frac{K_{sp}}{[\text{HC}_4\text{H}_4\text{O}_6^-]} = \frac{4.74 \times 10^{-4}}{0.0316} = 0.015 \text{ M.}$$

2. Ans (369.00)

$$1.8 \times 10^{-5} = \frac{[\text{CH}_3\text{COO}^-] \times 4 \times 10^{-4}}{0.4}$$

3. Ans (100.00)

$$W = -P (V_{\text{aragonite}} - V_{\text{calcite}})$$

$$= - \left(100 \times 10^5 \frac{\text{N}}{\text{m}^2} \right) \times \left(\frac{100}{2.5} - \frac{100}{2} \right) \times 10^{-6} \text{ m}^3$$

$$= +100 \text{ J.}$$

4. Ans (16.00)

$$10^{20} = \frac{0.01}{(2.5 \times 10^{-7})^2 \times [\text{Ag}^+]}$$

5. Ans (375.00)

$$P_1 = 2 \text{ bar, } T = 273 \text{ K, } P_2 = 4 \text{ bar}$$

$$PV^{-3/2} = \text{constant} \quad x = -3/2$$

$$\frac{\Delta U}{W} = \frac{n C_{vm} \Delta T}{\frac{n R (T_2 - T_1)}{(x-1)}} = \frac{C_{vm}(x-1)}{R} = \frac{3}{2} \left[-\frac{3}{2} - 1 \right]$$

$$= \frac{3}{2} \left[-\frac{3}{2} - 1 \right]$$

$$= -\frac{15}{4} = -3.75 \times 100 = 375$$

6. Ans (750.00)

$$\Delta H = E_{af} - E_{ab}$$

$$E_b = E_f + RT$$

$$A_f = 2A_b$$

$$K = 2 \times e^{-RT/RT} = \frac{2}{e}$$

$$\ln K = -0.3$$

$$-\frac{\Delta G^\circ}{RT} = -0.3$$

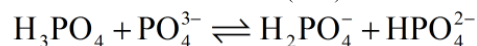
$$\text{or, } \Delta G^\circ = (-0.3) \times (-2500) = 750 \text{ J mol}^{-1}$$

PART-2 : CHEMISTRY

SECTION-III

1. Ans (3)

$$\begin{array}{ll} n_{\text{H}_3\text{PO}_4} = 150 \text{ mmol} & n_{\text{PO}_4^{3-}} = 50 \\ (\text{Acid}) & (\text{Base}) \end{array}$$



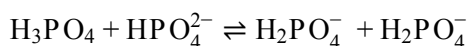
$$\text{initial } 150 \quad 50$$

$$\text{final } 100 \quad - \quad 50 \quad 50$$

(H_3PO_4 & H_2PO_4^-) and

(H_2PO_4^- & HPO_4^{2-}) are conjugates

So, no reaction



$$100 \quad 50$$

$$50 \quad - \quad 50 \quad 50$$

So finally we have 2000 ml solution of H_3PO_4 (50 m mole) and H_2PO_4^- (150 m mole) a buffer solution

$$\text{pH} = \text{pKa} + \log \frac{b}{a}$$

$$\text{pH} = 2.5 + \log 3 = 3$$

2. Ans (1)

Let heat involved in step 5 is Q_5

$$q_{\text{total}} = -W_{\text{total}}$$

$$500 + 800 + Q_5 = 700$$

$$Q_5 = -600 \text{ J}$$

Since $\Delta S_{\text{total}} = 0$

$$\frac{500}{250} + \frac{800}{200} - \frac{600}{T_5} = 0$$

$$2 + 4 - \frac{600}{T_5} = 0$$

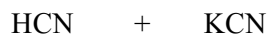
$$T_5 = 100 \text{ K}$$

3. Ans (6)

$$q = W = \Delta U = \Delta H = \Delta S_{\text{surr}} = 0$$

4. Ans (2)

Formation of buffer solution occurs.



2M, 10mL 5M, V mL

$$[\text{HCN}] = \frac{2 \times 10}{10 + V} \text{M}$$

$$[\text{KCN}] = \frac{5 \times V}{10 + V}$$

$$\text{pH} = \text{pK}_a + \log \left(\frac{[\text{KCN}]}{[\text{HCN}]} \right)$$

$$9 = 10 - \log 5 + \log \left(\frac{5V}{20} \right)$$

$$\log 5 - 1 = \log \left(\frac{5V}{20} \right)$$

$$\log \left(\frac{5}{10} \right) = \log \left(\frac{5V}{20} \right)$$

$$\frac{1}{2} = \frac{5V}{20}$$

$$V = 2 \text{ mL}$$

5. **Ans (2)**

$$\Delta S_{\text{sys}} = 0 \Rightarrow \text{Adiabatic process}$$

$$T.V^{\gamma-1} = \text{constant} \Rightarrow T.V^{5/3-1}$$

$$= \text{constant} \Rightarrow T^3 V^2 = \text{constant}.$$

6. **Ans (4)**



$$t = 0 \quad 5 \times 10^{-3} \quad 0.01$$

$$t = \infty \quad 5 \times 10^{-3} - x \quad 0.01 \times 2x \quad x \simeq 5 \times 10^{-7}$$

$$\frac{5 \times 10^{-3}}{(5 \times 10^{-3} - x)^3 \times 4} = 10^{22}$$

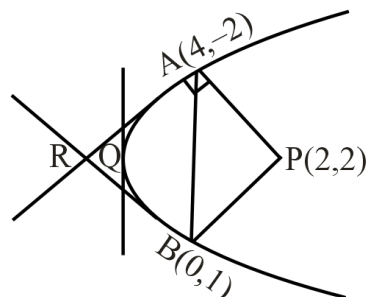
$$(5 \times 10^{-3} - x)^3 = 1.5 \times 10^{-25}$$

$$\Rightarrow [\text{Ag}^+] = 5 \times 10^{-3} - x = 5 \times 10^{-9}$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (B,D)



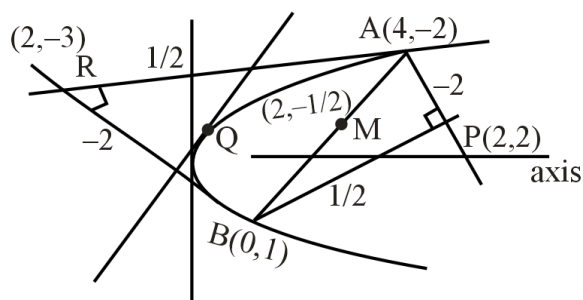
Observe that $AP \perp PB$

$$m_{AP} = -2; m_{PB} = \frac{1}{2}$$

So, ARBP is a rectangle

$\Rightarrow$ AB is a focal chord.

Hence we can find out $R(2, -3)$



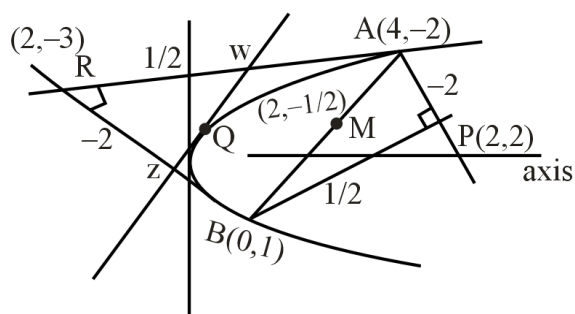
Say M $\left(2, -\frac{1}{2}\right)$ is mid point

of AB then RMP is parallel to the axis

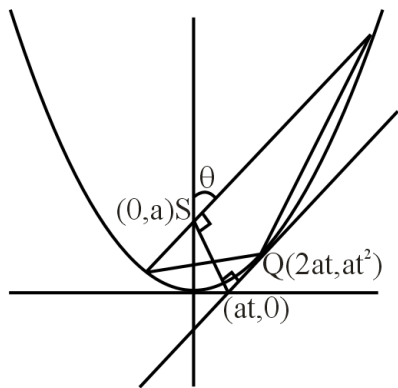
$\therefore$ Slope of axis is not defined

$\therefore$ Slope of directrix is zero which will pass

through $R(2, -3) \Rightarrow y = -3$ is directrix



Since directrix is horizontal so parabola is of the form $x^2 = 4ay$.



AB & tangent at Q are parallel.

So, distance of focus from tangent is a $\sqrt{1+t^2}$

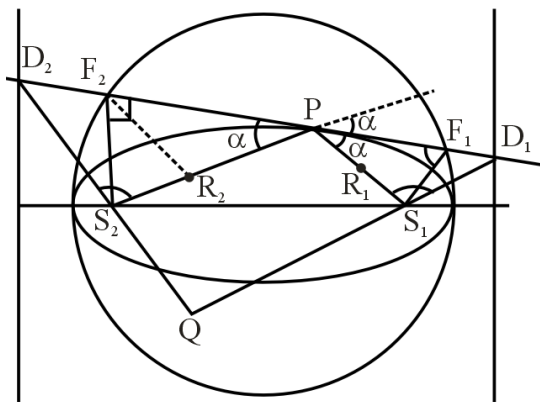
Slope of tangent = $t = m_{AB}$

$$\Rightarrow t = \frac{-3}{4} = \cot \theta$$

$$\text{and } 4a \operatorname{cosec}^2 \theta = 5 \Rightarrow a = \frac{4}{5}$$

$$\therefore \text{Area} = \frac{1}{2} \times AB \times a\sqrt{1+t^2} = \frac{5}{2}$$

2. Ans (A,C)



Let $\angle D_2PS_2 = \angle D_1PS_1 = \alpha$

{ $\therefore$ tangent bisects angle S_1PS_2 externally}

Then $PF_2 = PS_2 \cos \alpha$, $PF_1 = PS_1 \cos \alpha$

$$\Rightarrow F_1F_2 = PF_1 + PF_2 = (PS_1 + PS_2) \cos \alpha = a$$

$$\Rightarrow 2a \cos \alpha = a \Rightarrow \cos \alpha = \frac{1}{2} \Rightarrow \alpha = 60^\circ$$

$$\angle PD_1S_1 = 90^\circ - 60^\circ = 30^\circ$$

$$\Rightarrow \Delta D_1D_2Q \text{ is isosceles} \Rightarrow D_1Q = D_2Q$$

3. Ans (B,C)

$$\therefore \frac{1}{e^2} + \frac{1}{f^2(e)} = 1 \Rightarrow \frac{1}{f^2(e)} = 1 - \frac{1}{e^2} = \frac{e^2 - 1}{e^2}$$

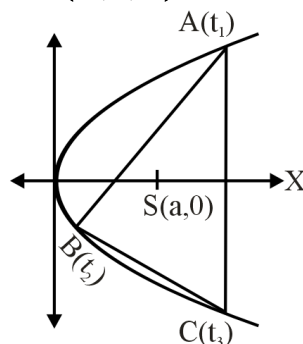
$$\Rightarrow f^2(e) = \frac{e^2}{e^2 - 1} \Rightarrow f(e) = \frac{e}{\sqrt{e^2 - 1}}$$

$$\therefore f(f(e)) = \frac{f(e)}{\sqrt{f^2(e) - 1}} = \frac{\frac{e}{\sqrt{e^2 - 1}}}{\sqrt{\frac{e^2}{e^2 - 1} - 1}} = e$$

$$\therefore f(f(f(e))) = f(e)$$

$$\therefore f(f(f(\dots f(e)\dots))) = \begin{cases} e & ; \text{ if } n \text{ is even} \\ \frac{e}{\sqrt{e^2 - 1}} & ; \text{ if } n \text{ is odd} \end{cases}$$

4. Ans (A,B,C)



$$\frac{2}{t_1 + t_2} \times \frac{2at_3}{a(t_3^2 - 1)} = -1$$

$$\frac{2}{t_1 + t_2} = \frac{1 - t_3^2}{2t_3}$$

$$t_1 + t_2 = \frac{4t_3}{1 - t_3^2}$$

$$t_1 + t_2 + t_3 = \frac{4t_3}{1 - t_3^2} + t_3$$

$$= \frac{4t_3 + t_3 - t_3^3}{1 - t_3^2} = \frac{5t_3 - t_3^3}{1 - t_3^2}$$

$$\text{If } t_1 + t_2 + t_3 = u \text{ then } u = \frac{5t_3 - t_3^3}{1 - t_3^2} \dots (1)$$

The Equation (1) suggests that t_3 is a root of

$$t^3 - ut^2 - 5t + u = 0 \dots (2)$$

by symmetry t_1 and t_2 are also the root of (2).

Hence

$$t_1 + t_2 + t_3 = u$$

$$\text{and } t_1t_2t_3 = -u$$

$$\text{Hence } t_1 + t_2 + t_3 + t_1t_2t_3 = 0$$

$$\text{also } \Sigma t_1t_2 = -5$$

$$\text{Hence } |t_1 + t_2 + t_3 + t_1t_2t_3 + \Sigma t_1t_2| = 5$$

5. Ans (B,D)

(A) Equation of hyperbola is $(x + 2y + 3)$

$$(3x + 4y + 5) + k = 0$$

$\therefore$ It passes through $(1, -1) \therefore k = -8$

$\therefore$ equation of conjugate hyperbola

– equation of asymptotes

= equation of asymptotes

– equation of hyperbola = + 8

$\Rightarrow$ Equation of conjugate hyperbola is :

$$(x + 2y + 3)(3x + 4y + 5) + 8 = 0$$

$$\Rightarrow 3x^2 + 10xy + 8y^2 + 14x + 22y + 23 = 0$$

$$\therefore \lambda = 14, \mu = 23$$

(B) $\therefore 3x + 4y = 0$ and $4x - 3y = 0$ are mutually perpendicular lines,

$$\text{then } \frac{\left(\frac{3x+4y}{5}\right)^2}{9} + \frac{\left(\frac{4x-3y}{5}\right)^2}{4} = 1$$

Vertices are points of intersection of

$$\frac{3x+4y}{5} = \pm 3 \text{ and } \frac{4x-3y}{5} = 0$$

$$\therefore \text{Vertices} = \left(\frac{9}{5}, \frac{12}{5}\right) \text{ and } \left(-\frac{9}{5}, -\frac{12}{5}\right)$$

$\therefore$ Given point is vertex

$\therefore$ minimum distance = $\lambda = 0$;

maximum distance = $\mu = 6$

(C) Q lies on directrix, which is image of $(1, 1)$

with respect to line $x + y = 7$

$$\Rightarrow Q \equiv (6, 6) \text{ PQ} \parallel \text{axis}$$

$$m_{PQ} = \frac{2}{3} \Rightarrow \text{axis } 2x - 3y + 1 = 0$$

Slope of the directrix is $-\frac{3}{2}$, which passes through $(6, 6)$

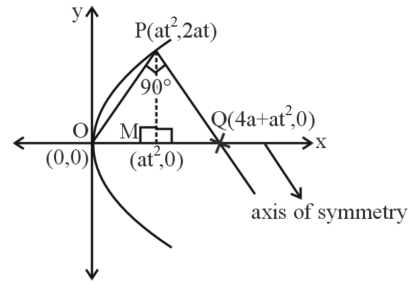
$$(D) (x - 1)^2 + (y - 3)^2 = \left(\frac{5x - 12y + 19}{13}\right)^2$$

Focus is $(1, 3)$ and directrix is $5x - 12y + 19 = 0$

latus rectum = $2(\text{distance of focus from directrix})$

$$= 2 \cdot \left| \frac{5 - 36 + 19}{13} \right| = \frac{24}{13} = \frac{\lambda}{\mu} \text{ then } \lambda = 24, \mu = 13$$

6. Ans (A,B,C)



$$(A) m_{OP} = \frac{2at - 0}{at^2 - 0} = \frac{2}{t}$$

$$\text{Equation of PQ is } (y - 2at) = \frac{-t}{2}(x - at^2)$$

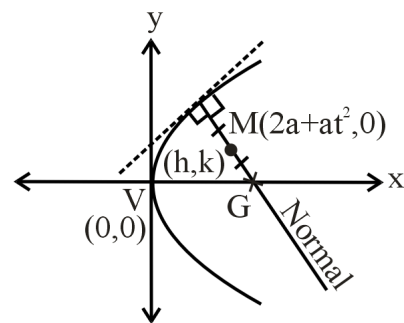
$$\text{Put } y = 0, x = 4a + at^2$$

Clearly projection of PQ on axis = $MQ = |4a$

$$+ at^2 - at^2| = 4a = \ell \text{ (Given)}$$

(B) Equation of normal at $P(at^2, 2at)$ is

$$y = -tx + 2at + at^3$$



$$\text{Put } y = 0, x = 2a + at^2$$

$$\therefore G(2a + at^2, 0)$$

$$\text{Now } 2h = 2a + 2at^2 \quad \dots(1)$$

$$\text{and } 2k = 2at \quad \dots(2)$$

$\therefore$ On eliminating t from (1) & (2)

we get locus of $M(h, k)$ is

$$y^2 = a(x - a) \Rightarrow \text{Latus rectum}$$

$$= a = \frac{\ell}{4} \text{ (Given)}$$

(C) We have $y^2 = \ell x$ (fixed parabola) ... (1)

Let variable parabola be $(y - k)^2 = -\ell(x - h)$... (2)

∴ On solving (1) and (2), we get

$$y^2 - 2ky + k^2 = -y^2 + \ell h$$

$$2y^2 - 2ky + (k^2 - \ell h) = 0$$

As (1) and (2) are tangent to each other, so

Discriminant = 0

$$\Rightarrow 4k^2 - 8(k^2 - \ell h) = 0 \Rightarrow k^2 - 2k^2 + 2\ell h = 0$$

$$\Rightarrow k^2 = 2\ell h$$

∴ Required locus is $y^2 = 2\ell x \Rightarrow$ Latus rectum = 2ℓ

(D) Clearly $2h = a + at^2$... (1)

and $2k = 2at$... (2)

∴ On eliminating t from (1) & (2),

we get locus of $M(h, k)$ is

$$y^2 = 2a \left(x - \frac{a}{2} \right) \Rightarrow \text{Latus rectum} = 2a = \frac{\ell}{2}$$

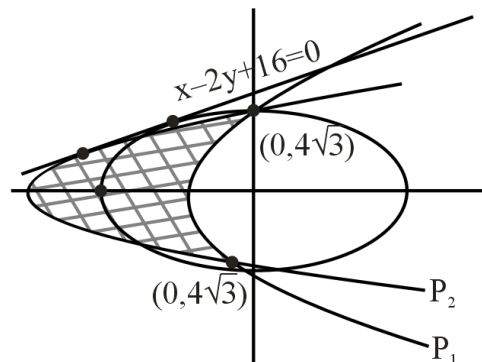
(Given)

PART-3 : MATHEMATICS

SECTION-II

1. Ans (42.66 or 42.67)

Equation of common tangent is



$$y = \frac{1}{2}x + 8 \quad \begin{cases} m = \frac{1}{2} \\ c = 8 \end{cases}$$

Let equation of $P_1 : y^2 = 4a(x - \alpha)$

and $P_2 : y^2 = 4b(x - \beta)$

both passes through $(0, \pm 4\sqrt{3})$

$$\Rightarrow \alpha = -\frac{12}{a}; \beta = -\frac{12}{b}$$

$$\Rightarrow P_1 : y^2 = 4a \left(x + \frac{12}{a} \right)$$

$$\Rightarrow P_2 : y^2 = 4b \left(x + \frac{12}{b} \right)$$

equation of tangents are

$$y = m \left(x + \frac{12}{a} \right) + \frac{a}{m}, y = m \left(x + \frac{12}{b} \right) + \frac{b}{m}$$

comparing with $y = \frac{1}{2}x + 8$

$$\Rightarrow m = \frac{1}{2}, \frac{6}{a} + 2a = 8, \frac{6}{b} + 2b = 8$$

$$\Rightarrow a = 3, b = 1$$

$$\Rightarrow \alpha = -4, \beta = -12$$

equation of both the parabola are

$$y^2 = 12(x + 4), y^2 = 4(x + 12)$$

$$\Rightarrow A = \int_{-4\sqrt{3}}^{4\sqrt{3}} \left[\left(\frac{y^2}{12} - 4 \right) - \left(\frac{y^2}{4} - 12 \right) \right] dy$$

$$= \frac{128}{\sqrt{3}}$$

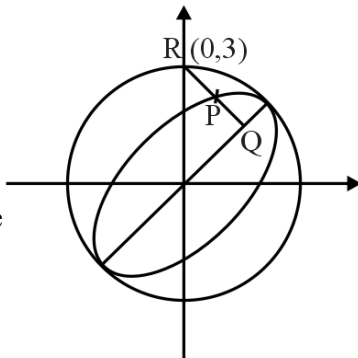
2. Ans (72.00)

Let length of semimajor axis of ellipse = a
and length of semimajor axis of ellipse = b

$$a = 3 \text{ and } RQ = \left| \frac{3.0 - 4.3}{5} \right| = \frac{12}{5}$$

$$\begin{aligned} \therefore \frac{PQ}{RQ} &= \frac{b}{a} \\ \Rightarrow \frac{b}{3} &= \frac{5}{12} \\ \Rightarrow b &= \frac{5}{4} \end{aligned}$$

Now, distance
between foci = $2ae$



$$\begin{aligned} d &= 2\sqrt{a^2 - b^2} = 2\sqrt{9 - \frac{25}{16}} \\ \Rightarrow \frac{288d^2}{119} &= 36 \end{aligned}$$

3. Ans (16.00)

Let $Y = y - 4$ and $X = x + 3$

$$Y^2 = 15X$$

Let $P(t_1)$, $Q(t_2)$ and $R(t_3)$

$$\therefore t_3 = -t_1 - \frac{2}{t_1} = -t_2 - \frac{2}{t_2}$$

$$\therefore t_1 t_2 = 2 \quad \dots(i)$$

$$t_1 + t_2 + t_3 = 0 \quad \dots(ii)$$

let (h, k) be centroid under new co-ordinate system

$$\begin{aligned} h &= \frac{a(t_1^2 + t_2^2 + t_3^2)}{3} = \frac{15}{12}(t_1^2 + t_2^2 + (t_1 + t_2)^2) \\ &= \frac{15}{6}(t_1^2 + t_2^2 + 2) = \frac{15}{6}(t_1^2 + \frac{4}{t_1^2} + 2) \end{aligned}$$

$$\therefore t_1^2 + \frac{4}{t_1^2} \geq 4$$

but equality is not taken

$$\therefore h > \frac{15}{6}(4 + 2)$$

$$h > 15 \text{ and } k = 2a(t_1 + t_2 + t_3) = 0$$

$$\alpha + 3 = h > 15 \quad \alpha > 12$$

$$\beta - 4 = k = 0$$

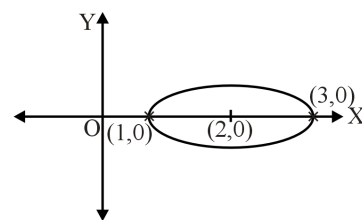
$$\beta = 4$$

$$\alpha + \beta > 16$$

largest value of $m = 16$

4. Ans (38.00)

$$\text{As } \vec{a} \cdot \vec{b} = 0$$



$$\Rightarrow (x^2 + 3) + 9y^2 - 4x = 0 \Rightarrow (x - 2)^2 + 9y^2 = 1$$

$$\Rightarrow \frac{(x - 2)^2}{1} + \frac{y^2}{1/9} = 1,$$

which is equation of an ellipse.

$$\text{Now, } (x - 2)^2 = 1 - 9y^2 \geq 0 \Rightarrow (3y + 1)(3y - 1) \leq 0$$

$$\Rightarrow -\frac{1}{3} \leq y \leq \frac{1}{3}$$

$$\text{Also, } y^2 = \frac{1 - (x - 2)^2}{9} \geq 0 \Rightarrow (x - 2)^2 - 1 \leq 0$$

$$\Rightarrow (x - 3)(x - 1) \leq 0$$

$$\text{So, } x \in [1, 3]; y \in \left[-\frac{1}{3}, \frac{1}{3}\right]$$

Clearly, $m = 3$ and $n = 1$.

$$\text{Hence, } (m^2 + n^2) = 9 + 1 = 10$$

Using family of circles, the equation of circle which is tangent to conic C at $M(3, 0)$ is

$$(x - 3)^2 + y^2 + \lambda(x - 3) = 0$$

As above circle passes through $(2, 0)$, so

$$1 + 0 - \lambda = 0 \Rightarrow \lambda = 1$$

$$\begin{aligned} \text{So, equation of circle is } (x - 3)^2 + y^2 + x - 3 &= 0 \Rightarrow x^2 \\ &+ y^2 - 5x + 6 = 0. \end{aligned}$$

$$\text{whose centre is } \left(\frac{5}{2}, 0\right) \text{ and radius} = \frac{1}{2}$$

So, required equation of director circle of above circle is

$$\left(x - \frac{5}{2}\right)^2 + y^2 = \frac{1}{2} \Rightarrow (2x - 5)^2 + 4y^2 = 2.$$

$$\text{i.e., } 4x^2 + 4y^2 - 20x + 23 = 0.$$

5. Ans (32.00)

(I) Let $y = mx + c$ is tangent to $x^2 + y^2$

$$= 4 \text{ and } \frac{x^2}{5} + \frac{y^2}{3} = 1$$

$$c^2 = 4(1 + m^2) = 5m^2 + 3$$

$$\Rightarrow m^2 = \frac{4-3}{5-4} = 1$$

$$\Rightarrow m^2 = 1 \text{ \& } c^2 = 8$$

x-coordinate of point of tangent on circle is

$$(1 + m^2)x^2 + 2mcx^2 + c^2 - 4 = 0$$

$$\Rightarrow 2x^2 - 2cx^2 + c^2 - 4 = 0$$

$$\Rightarrow 2x^2 - 2cx + 4 = 0 \Rightarrow x^2 - cx + 2 = 0$$

$$\Rightarrow x^2 - 2\sqrt{2}x + 2 = 0 \Rightarrow x = \sqrt{2}$$

$$\therefore A (\sqrt{2}, \sqrt{2})$$

Point of tangent of ellipse

$$B \left(\frac{5}{2\sqrt{2}}, \frac{3}{2\sqrt{2}} \right)$$

$$\text{Length of common tangent } AB = \frac{1}{2} = \frac{1}{L}$$

$$\therefore L = 2$$

(II) $y = -(5/2)x + 5 \Rightarrow m = 2/5 \Rightarrow a^2m^2 - b^2$

$$= 9.4/25 - 4 = (36 - 100)/25 < 0$$

Note that the slope of the tangent (2/5) is less than the slope of the asymptote which is 2/3 which is not possible.

(III) Normal at $P(3\sec\theta, 4\tan\theta)$ is

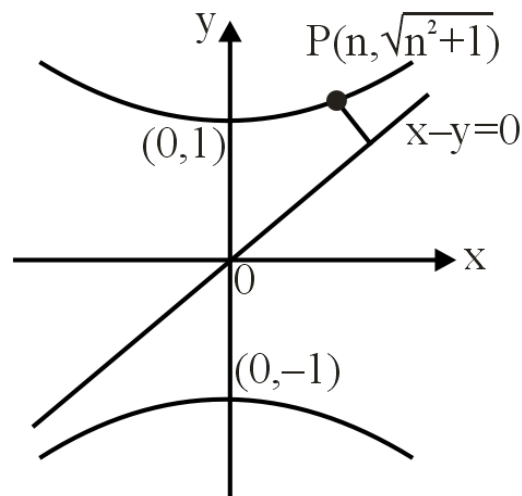
$$\frac{3x}{\sec\theta} + \frac{4y}{\tan\theta} = 25 \text{ passes through } (5,0)$$

$$\Rightarrow \sec\theta = \frac{15}{25} \text{ (not possible for } \theta \in (0, \frac{\pi}{2}))$$

Hence $\theta = 0 \Rightarrow$ Point of tangency is (3, 0) and

radius = 2

(IV) Curve is rectangular hyperbola.



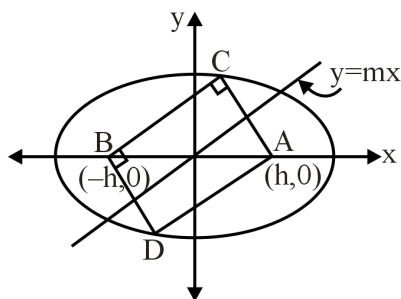
$$\text{perpendicular distance, } d_n = \left| \frac{n - \sqrt{n^2 + 1}}{\sqrt{2}} \right|$$

$$\lim_{n \rightarrow \infty} (n \cdot d_n) = \lim_{n \rightarrow \infty} \frac{n}{\sqrt{2}} \left| \left(\sqrt{n^2 + 1} - n \right) \right|$$

$$= \lim_{n \rightarrow \infty} \frac{n}{\sqrt{2}} \left| \frac{1}{\sqrt{n^2 + 1} + n} \right| = \frac{1}{2\sqrt{2}}$$

6. Ans (50.00)

Image of A(h, 0) in the line mirror $mx - y = 0$



$$\frac{x-h}{m} = \frac{y-0}{-1} = -2 \left(\frac{mh}{m^2+1} \right)$$

$$x = \frac{h(1-m^2)}{1+m^2}, y = \frac{2mh}{m^2+1}$$

$$\therefore C \equiv \left(\frac{h(1-m^2)}{1+m^2}, \frac{2mh}{m^2+1} \right)$$

$$\text{Similarly } D \equiv \left(\frac{-h(1-m^2)}{1+m^2}, \frac{-2mh}{m^2+1} \right)$$

$$AC = \sqrt{\left(h - \frac{h(1-m^2)}{1+m^2} \right)^2 + \frac{4m^2h^2}{(1+m^2)^2}}$$

$$= \sqrt{\frac{4m^4h^2}{(1+m^2)^2} + \frac{4m^2h^2}{(1+m^2)^2}} = \frac{2mh\sqrt{1+m^2}}{(1+m^2)}$$

$$AD = \sqrt{\left(h + \frac{h(1-m^2)}{1+m^2} \right)^2 + \frac{4m^2h^2}{(1+m^2)^2}}$$

$$= \sqrt{\frac{4h^2}{(1+m^2)^2} + \frac{4m^2h^2}{(1+m^2)^2}}$$

$$= \frac{2h}{(1+m^2)} \sqrt{1+m^2}$$

$$\therefore \text{Area} = AC \cdot AD = \frac{4mh^2(1+m^2)}{(1+m^2)^2}$$

$$= \frac{4h^2}{m + \frac{1}{m}}$$

$$\text{Area}|_{\max} = \frac{4 \times 25}{2} = 50$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (6)

An isosceles $\triangle ABC$ is inscribed in the circle $x^2 + y^2 = 16$

$$A \equiv (4\cos\theta, 4\sin\theta) \Rightarrow P \equiv (4\cos\theta, -3\sin\theta)$$

$$B \equiv (4\cos(\theta + \alpha), 4\sin(\theta + \alpha)) \Rightarrow Q \equiv (4\cos(\theta + \alpha), 3\sin(\theta + \alpha))$$

$$C \equiv (4\cos(\theta - \alpha), 4\sin(\theta - \alpha)) \Rightarrow R \equiv (4\cos(\theta - \alpha), 3\sin(\theta - \alpha))$$

As PQR is a right angled triangle

$$m_{PQ} \cdot m_{PR} = -1$$

$$\Rightarrow \frac{9}{16} \cdot \left[\frac{\sin\theta + \sin(\theta + \alpha)}{\cos\theta - \cos(\theta + \alpha)} \right] \left[\frac{\sin\theta + \sin(\theta - \alpha)}{\cos\theta - \cos(\theta - \alpha)} \right] = -1$$

$$\Rightarrow \frac{9}{16} \left[\frac{\sin\left(\theta + \frac{\alpha}{2}\right) \cos\left(\frac{\alpha}{2}\right)}{\sin\left(\frac{\alpha}{2}\right) \sin\left(\theta + \frac{\alpha}{2}\right)} \cdot \frac{\sin\left(\theta - \frac{\alpha}{2}\right) \cos\left(\frac{\alpha}{2}\right)}{\sin\left(-\frac{\alpha}{2}\right) \sin\left(\theta - \frac{\alpha}{2}\right)} \right] = -1$$

$$\Rightarrow \frac{9}{16} \left(-\cot^2 \frac{\alpha}{2} \right) = -1$$

$$\Rightarrow \cot^2 \frac{\alpha}{2} = \frac{16}{9}$$

$$\Rightarrow \tan \frac{\alpha}{2} = \frac{3}{4}$$

angles of $\triangle ABC$ will be $\frac{\alpha}{2}, \frac{\alpha}{2}, \pi - \alpha$

so angles are $\tan^{-1} \frac{3}{4}, \tan^{-1} \frac{3}{4},$

$$\pi - 2\tan^{-1} \frac{3}{4} \Rightarrow \tan\theta = \frac{3}{4}$$

$$\Rightarrow 4\tan\theta + 3 = 6$$

2. Ans (4)

Tangent to the Parabola

$$y = mx - \frac{a}{4m}$$

$$(1, 0) \Rightarrow a = 4m^2$$

(1, 0) lies on the directrix of the hyperbola

$$\frac{2}{e} = 1 \Rightarrow e = 2$$

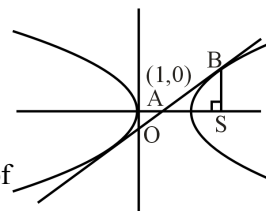
$$\Rightarrow 1 + \frac{b^2}{4} \Rightarrow b^2 = 12$$

Tangent to hyperbola : $y = mx \pm \sqrt{4m^2 - b^2}$

$$\frac{a^2}{16m^2} = 4m^2 - b^2 \Rightarrow m^2 = 4m^2 - b^2 \Rightarrow 3m^2 = 12$$

$$\Rightarrow m^2 = 4$$

$$\Rightarrow a = 16 \Rightarrow a - b^2 = 16 - 12 = 4$$



3. Ans (6)

$$\frac{x^2}{16} + \frac{y^2}{1} = 1$$

$$a^2 = 16, b^2 = 1$$

$$a = 4, b = 1$$

let $A(x_1, y_1)$ and $B(x_2, y_2)$ then P, Q will be

$$P\left(x_1, \frac{y_1}{4}\right) \text{ and } Q\left(x_2, \frac{y_2}{4}\right)$$

$$\text{mid point of AB will be } \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

$$\text{mid point of PQ will be } \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{8}\right) = (h, k)$$

$$\Rightarrow \text{mid point of AB will be } (h, 4k)$$

let point $S(t, t+8)$ be a point on $y = x + 8$

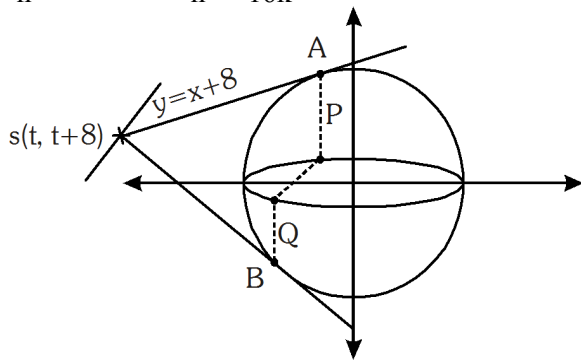
AB is chord of contact of $x^2 + y^2 = 16$

$$\Rightarrow xt + y(t+8) = 16 \quad \dots(i)$$

AB as chord whose mid point is $(h, 4k)$

$$xh + y(4k) = h^2 + 16k^2 \quad \dots(ii)$$

$$\frac{t}{h} = \frac{t+8}{4k} = \frac{16}{h^2 + 16k^2}$$



$$\Rightarrow t = \frac{16h}{h^2 + 16k^2} \text{ and } t+8 = \frac{64k}{h^2 + 16k^2}$$

$$\Rightarrow \frac{16h}{h^2 + 16k^2} + 8 = \frac{64k}{h^2 + 16k^2}$$

$$\Rightarrow 2h + (h^2 + 16k^2) = 8k$$

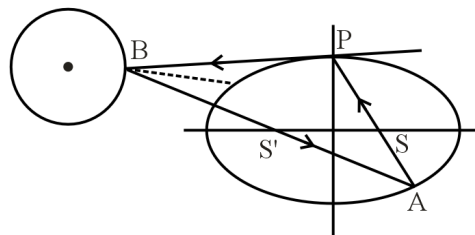
$$\Rightarrow x^2 + 16y^2 + 2x - 8y = 0$$

$$x^2 + \alpha y^2 + \beta x + \gamma y = 0$$

$$\Rightarrow \alpha = 16, \beta = 2, \gamma = -8$$

$$\Rightarrow \alpha - \beta + \gamma = 16$$

4. Ans (5)



Using reflection property of ellipse equation of reflected ray SP is

$$4x + 3y = 12 \quad \dots(1)$$

Solving equation (1) with ellipse, we get

$$A \equiv \left(\frac{75}{17}, -\frac{32}{17}\right)$$

Equation of reflected ray BS' is

$$16x + 63y + 48 = 0 \quad \dots(2)$$

Equation of incident ray PB is

$$y = 4 \quad \dots(3)$$

Equation of bisector of (2) and (3) with negative slope gives normal to circle at 3 which is

$$4x + 32y - 53 = 0$$

Point B is $\left(-\frac{75}{4}, 4\right)$, using parametric equation of line ordinate of centre is 5

5. Ans (7)

Equation of tangent to the parabola $y^2 = 4ax$ is

$$y = mx + \frac{a}{m} \text{ which touches the circle } x^2 + y^2 = \frac{a^2}{2}$$

$$\therefore \left(\frac{a}{m}\right)^2 = \left(\frac{a}{\sqrt{2}}\right)^2 (1 + m^2) \Rightarrow m^4 + m^2 - 2 = 0$$

$$\Rightarrow (m^2 + 2)(m^2 - 1) = 0$$

$$m = \pm 1$$

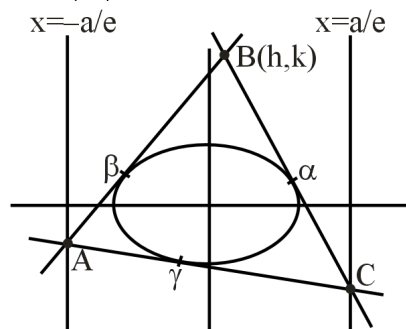
$\therefore$ Equation of tangents are $y = x + a$ and $y = -x - a$

and also clear that $P(-a, 0)$ so equation of chord of contact to the circle and parabola are $x = \frac{-a}{2}$ and $x = a$ respectively.

$$\therefore \text{Area of trapezium ABCD} = \frac{1}{2}(AB + CD) \cdot$$

$$RS = \frac{1}{2}(a + 4a) \cdot \frac{3a}{2} = \frac{15a^2}{4} \text{ square units}$$

6. Ans (2)



$$\text{Given } \frac{a \cos\left(\frac{\beta+\gamma}{2}\right)}{\cos\left(\frac{\beta-\gamma}{2}\right)} = -\frac{a}{e}$$

$$\Rightarrow \frac{\cos \frac{\beta}{2} \cos \frac{\gamma}{2} - \sin \frac{\beta}{2} \sin \frac{\gamma}{2}}{\cos \frac{\beta}{2} \cos \frac{\gamma}{2} + \sin \frac{\beta}{2} \sin \frac{\gamma}{2}} = -\frac{1}{e}$$

$$\Rightarrow \frac{1 - \tan \frac{\beta}{2} \tan \frac{\gamma}{2}}{1 + \tan \frac{\beta}{2} \tan \frac{\gamma}{2}} = -\frac{1}{e}$$

$$\& \frac{a \cos \frac{\alpha+\gamma}{2}}{\cos \frac{\alpha-\gamma}{2}} = \frac{a}{e} \Rightarrow \frac{1 - \tan \frac{\alpha}{2} \tan \frac{\gamma}{2}}{1 + \tan \frac{\alpha}{2} \tan \frac{\gamma}{2}} = \frac{1}{e}$$

to find the locus of

$$h = \frac{a \cos\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)} ; K = \frac{b \sin\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)}$$

$$h = \frac{a \left(1 - \tan \frac{\alpha}{2} \tan \frac{\beta}{2}\right)}{1 + \tan \frac{\alpha}{2} \tan \frac{\beta}{2}}$$

$$K = b \cdot \frac{\tan \frac{\alpha}{2} + \tan \frac{\beta}{2}}{1 + \tan \frac{\alpha}{2} \tan \frac{\beta}{2}}$$

$$\text{Let } \tan \frac{\alpha}{2} = A, \tan \frac{\beta}{2} = B, \tan \frac{\gamma}{2} = C$$

$$\text{i.e. } \frac{1 - BC}{1 + BC} = -\frac{1}{e} \quad \dots(1) \quad \&$$

$$\frac{1 - AC}{1 + AC} = \frac{1}{e} \quad \dots(2)$$

$$h = a \frac{1 - AB}{1 + AB} \quad \dots(3) \quad \&$$

$$k = b \frac{A + B}{1 + AB} \quad \dots(4)$$

$$\text{using C \& D in (1)} \quad \frac{2BC}{2} = \frac{e+1}{e-1}$$

$$\text{using C \& D in (2)} \quad AC = \frac{e-1}{e+1}$$

$$\Rightarrow \text{divide } \frac{B}{A} = \left(\frac{e+1}{e-1}\right)^2 \quad \dots(5)$$

$$\text{using C \& D in (3)} \quad AB = \frac{a-h}{a+h} \quad \dots(6)$$

$$\text{Multiply (5) \& (6)} \quad B^2 = \left(\frac{e+1}{e-1}\right)^2 \frac{a-h}{a+h}$$

$$B = \frac{e+1}{e-1} \sqrt{\frac{a-h}{a+h}} ;$$

$$A = \frac{a-h}{a+h} \div B = \sqrt{\frac{a-h}{a+h}} \times \frac{e+1}{e-1}$$

hence put these values of A & B in (4)

$$\frac{K}{b} = \frac{\sqrt{\frac{a-h}{a+h}} \cdot \frac{e-1}{e+1} + \frac{e+1}{e-1} \cdot \sqrt{\frac{a-h}{a+h}}}{1 + \frac{a-h}{a+h}}$$

$$= \frac{\sqrt{\frac{a-h}{a+h}} \left[\frac{(e-1)^2 + (e+1)^2}{e^2 - 1} \right]}{\frac{2a}{a+h}}$$

$$\frac{K}{b} = \frac{\sqrt{a^2 - h^2} \cdot 2 \cdot (e^2 + 1)}{(e^2 - 1) \cdot 2a} \quad e^2 = 1 - \frac{b^2}{a^2}$$

$$-Kb = a\sqrt{a^2 - h^2}(1 + e^2)$$

$$\text{hence } y^2 b^2 = a^2(1 + e^2)(a^2 - x^2)$$